feature/*: Branches for individual features created from *develop*.
release/*: Branches for preparing a new release created from *develop*.
hotfix/*: Branches for urgent fixes created from *main*.

Process:

- Develop features on *feature* branches.
- Merge *feature* branches into *develop*.
- Create a *release* branch from *develop* to finalise and prepare for a release.
- Merge *release* branch into both *main* and *develop*.
- Create a *hotfix* branch from *main* for urgent fixes and merge it into both *main* and *develop*.

Use case: Best suited for projects with a regular release cycle and where a structured approach to feature development, releases, and hotfixes is needed.

- **Forking workflow**

Forking workflow is commonly used in open source projects. Each contributor forks the repository, works on their own copy, and then submits pull requests to the original repository.

Process:

- Create a personal copy of the repository (fork) on a hosting service like GitHub.
- Clone the forked repository to your local machine.
- Make changes and commit them to your forked repository.
- Push the changes to your fork and submit a pull request to the original repository.

Use case: Ideal for open source projects or projects where many contributors are involved, especially when contributors do not have direct write access to the main repository.

Installing and configuring Git

To install and configure Git, start by downloading the appropriate installer for your operating system (Windows, macOS, Linux (Debian/Ubuntu)).

For Windows, visit *git-scm.com*, download the installer, and follow the on-screen instructions with the default settings.

On macOS, open *Terminal* and install Git using Homebrew by running *brew install git.*

For Linux users, particularly those on Debian or Ubuntu, open *Terminal* and execute *sudo apt update* followed by *sudo apt install git* to install Git.

Once Git is installed, you need to configure it with your user information. Open your terminal and set your username by running *git config --global user.name "Your Name"*, and then set your email with *git config --global user.email "youremail@example.com"*.

To ensure that Git has been installed correctly, check the version by running *git --version* and view your configuration with *git config --list*. This will prepare Git for version control tasks and ensure it recognises your commits with the correct user information.

How to create a Git repository

Creating a Git repository involves initialising a new repository in your project directory or cloning an existing repository. Here's a step-by-step guide on how to create a Git repository.

Step 1: Ensure Git is installed on your system. You can download it from *git-scm.com.*

- Open your terminal or command prompt.
- Navigate to your project directory using the *cd* command. For example:

```
cd path/to/your/project
```

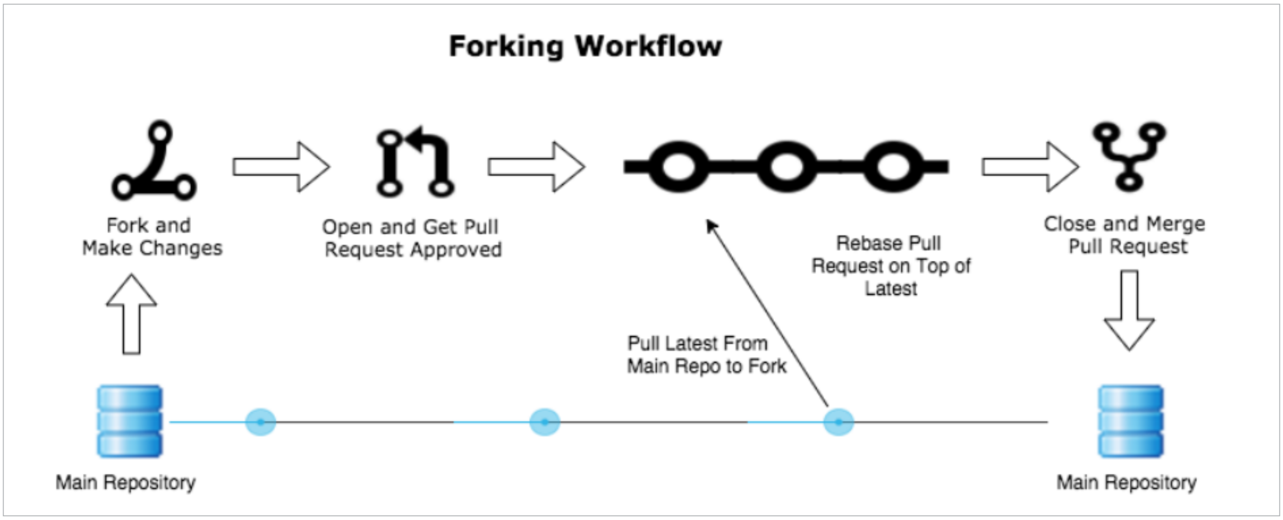

Figure 12: Forking workflow